

5.6 Portable Media Players

5.6.1 Audio Formats

All PMPs invariably support MP3 audio, so that music can be easily swapped from your computer to your PMP and back, as MP3 is one of the most popular formats with a huge volume of existing media. Other formats supported are Microsoft's WMA and also the open source Ogg Vorbis. WMA reduces the size of MP3 files by almost a factor of two, but due its proprietary nature, costs could increase. Ogg is technically superior in terms of compression, but lack of availability of media can be a problem.

AAC (Advanced Audio Coding) is a lossy compression and coding scheme for audio, and is touted to be the successor to MP3. AAC delivers better quality than MP3 at the same bitrate. AAC is the standard format used on the iPod and the iTunes store, and in the PlayStation 3. Other music players that support AAC include the Creative Zen and Microsoft's Zune.

5.6.2 Bitrate

The amount of data per second in a media file is known as the bitrate, and is expressed in kilobits per second (kbps). The higher the bitrate, the higher would be the quality, and more storage space would be needed. Bitrate can also be variable, in which the rate at which data is packed in is varied over time. This means that more data can be stored for complex segments of media files, while lesser data is stored in simpler segments.

5.6.3 Connectors

The main types of ports in a PMP are the audio-out ports, which are used to pipe music to speakers or headphones. Some PMPs offer a TV-out port, which lets you hook it up to a large screen. USB and Mini USB ports are standard offerings for transferring media, along with power ports. iPods have a FireWire port, which allows simultaneous charging and data transfer. The FireWire port can also be used to attach to external speakers and the ubiquitous iPod dock.